Dr. Rishal Mohammed, BDS, MDS, MFD RCSI

PhD Scholar | Oral Pathology | AI in Oral Health Research
Email:rishaljafar@gmail.com| +1 263 556-9402

ACADEMIC PROFILE

Research-oriented Oral Pathologist with experience in academics and clinical dentistry, with a focus on artificial intelligence in oral medicine. Strong background in oral pathology, public health data analysis, and evidence synthesis. Experienced in interdisciplinary research collaborations, student mentorship, and scholarly publishing. A PhD Scholar at McGill University, Canada with a focus on oral health research and AI-enabled healthcare

RESEARCH INTERESTS

- Artificial Intelligence in Oral Medicine and Diagnostics
- Oral Pathology and Oral Cancer Research
- Public Health Dentistry
- Evidence-Based Dentistry (Systematic Reviews & Meta-Analysis, network meta-analysis, Scoping review)
- Digital Health, mHealth, and Clinical Decision Support Systems

EDUCATION

Master of Dental Surgery (MDS) – Oral Pathology & Microbiology | Government Dental College, Kottayam, India | *2019*

Bachelor of Dental Surgery (BDS) | Kerala University, India | *2014*

MFD RCSI (Member of the Faculty of Dentistry) | Royal College of Surgeons in Ireland

RESEARCH EXPERIENCE

Research Assistant – Artificial Intelligence in Oral Medicine

Qatar University, College of Dental Medicine | *Mar 2024 – Sep 2025*

- Contributed to development of AI ensemble models (BCCN + BLR) for oral lesion analysis through large-scale clinical data annotation and curation.
- Supported design and deployment of AI-based validation and educational web platforms for academic and clinical users.
- Conducted scoping reviews and evidence synthesis on AI applications in oral healthcare, including treatment networks, mobile health usability, and regulatory frameworks.

- Designed and administered Knowledge, Attitude, and Practice (KAP) surveys evaluating adoption of AI in oral cancer diagnosis.
- Collaborated with international academic and research teams as part of global consortia.

SCHOLARSHIPS AND AWARDS

- Doctoral Research Award: Canada Impact+ Research Training Awards – CIHR 2026
- Summer Institutes In Health Research Methodologies, McGill. NCOHR in-person attendance support

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ACADEMIC & TEACHING EXPERIENCE

Senior Resident – Department of Public Health Dentistry

Government Dental College, Calicut, India | *Oct 2019 – Nov 2020*

- Delivered lectures and practical training for undergraduate and postgraduate dental students in the department of Oral Pathology and Public health dentistry.
- Community Outreach: Organized and supervised dental camps in schools, colleges, and tribal areas and Habit cessation programs.
- Supervised student research projects and supported manuscript preparation.

Junior Resident – Department of Oral Pathology & Microbiology

Government Dental College, Kottayam, India | *May 2016 – May 2019*

- Provided diagnostic oral pathology services, including slide reporting and clinical-pathological correlation.
- Participated in departmental research, seminars, and postgraduate academic activities.

CLINICAL EXPERIENCE (SUMMARY)

Consultant Oral Pathologist and General Dentist with experience in private and institutional dental practice in India.

RESEARCH METHODS & TECHNICAL SKILLS

- Oral Pathology Diagnostics (Histopathology, FNAC, Biopsy)
- Artificial Intelligence in Healthcare Research
- Cancer Genomic Databases (GDC, COSMIC)
- Public Health Databases (QBB, NHANES, NDI)
- Statistical Analysis: Python, R, STATA

- Systematic Reviews, Meta-Analysis, Scoping Reviews
- Survey Design and Validation (KAP Studies)
- Research Protocol Development & Grant Writing

PUBLICATIONS

Published Articles

- Niher Tabassum Snigdha, Shazia Fathima J.H, Kumar Krishnan, Santosh R Patil, Rishal Mohammed Pathuthara, Subtype-Dependent Progression Trajectories in Oral Submucous Fibrosis: A Longitudinal Prognostic Modelling Study with Clinical Decision Framework APJCP-2605-13282
- Fathima JH, Priya MS, Pathuthara RM, Rajagopal P, Murali R, Veeraraghavan VP, *et al.* Expression profile of miR-99a and miR-218 across oral submucous fibrosis and oral squamous cell carcinoma. J Oral Maxillofac Pathol 2026;30:132-8.
- Gupta, Sameer[1]; Mehra, Pravesh[1]; Nandy, Bishan[2]; Mohammed, P Rishal[3]; Pradhan, Prasanti K.[4]; Duseja, Shilpa[5]; Sharma, Nirvi[6]; Pujari, Priyanka R.[7]. Blockchain in Health Care, Evolutionary or Revolutionary – An Evidence-Based Review. Journal of Pharmacy and Bioallied Sciences 17(Suppl 1):p S63-S65, May 2025. | DOI: 10.4103/jpbs.jpbs_672_24
- Mohammed, P Rishal1; Dash, Mitali2; Pati, Seema3; Yaqoob, Ashfaq4; Sanda, Sailaja5; Nelson, Anish6; Tiwari, Heena D.7. Genetic Markers as Predictors of Prognosis and Treatment Response in Patients with Oral Cancer: A Prospective Study. Journal of Pharmacy and Bioallied Sciences 17(Suppl 1):p S439-S441, May 2025. | DOI: 10.4103/jpbs.jpbs_1447_24
- Beena VT, Kavitha A, et al. Sclerosing Polycystic Adenosis: A Rare and Unusual Lesion of the Salivary Gland. Oral Maxillofac Patho J 2019;10(2):95–98.
- Beena VT, Mohammed R, Paul S, Stephen MM, Nair C, Mohan AP. Ameloglyphics: The Tooth Signature. Oral Maxillofac Pathol J 2018;9(2):70-75.
- VT Beena, Nair C, Cherian LM, Indu M, Paul S, Mohammed R. A Rare Tumor of the Anterior Maxilla Masquerading as a Periapical Cyst: Case Report of Squamous Odontogenic Tumor. Oral Maxillofac Pathol J 2019;10(1):51-55.
- Beena VT, Revathy K, Cherian LM, Indu M, Mohammed R, Remya K. Two Case Reports of Actinomycosis: The Mysterious Ray Fungi with Spurious Clinical Diagnosis. Oral Maxillofac Pathol J 2019;10(1):25-29.
- Rishal Mohammed P, Jasmine Jose, Dlishad Banu and Dhanya S Morphometric Analysis of Mandible for Sex Determination; Acta Scientific Dental Sciences 2022; 6(10). DOI: 10.31080/ASDS.2022.06.1475
- Beena, Valappil T., Sabu Paul, R Mohammed, Pradeesh Sathyan, A. P. Kavitha and Revathy Kala. "Don't let the Clinical Diagnosis Prejudice the Histopathological Reporting: A Case of Suture Granuloma." (2020). DOI:10.38110/ijohmr.2020.v06i05.006
- Dubey, S., Rishal, M. P., Singh, H., Nikitha, A. V., Lavanya, C., Anand, P., & Rajesh, D. (2022). Evaluation of oral manifestations in COVID patients an original research. *International Journal of Health Sciences*, *6*(S5), 10252–10259. https://doi.org/10.53730/ijhs.v6nS5.12152

Manuscripts Under Journal's Consideration [under review]

- Nanocarrier-Mediated miRNA Delivery in Oral Squamous Cell Carcinoma: A Comprehensive Review of Delivery Architecture, Preclinical Efficacy and Translational Barriers – Discover Nano

- Virtual Microscopy in Oral Pathology Education: A Systematic Search with Narrative Synthesis of Pedagogical Efficacy, Diagnostic Concordance and Emerging AI Applications – The Saudi Dental Journal
- Mapping Network Meta-Analysis Evidence for the Treatment of Oral Lesions: A Scoping Review (TRIPLEO-D-26-01749)
- Age-Related Changes in Electrochemistry and Ionic Balance of Human Saliva: A Cross-Sectional Study – *TRIPLE O)*
- Association of Salivary Electrochemistry and Ionic Balance with Tooth Shade and Mechanical Property – *Journal of prosthodontics*
- Artificial Intelligence for Oral Cancer Screening: a Multi-Center and Multi-National Study, Followed by Comparative Analysis Against Health-care Professionals *The Lancet Digital Health*
- Artificial Intelligence Models for Predicting Recurrence of Odontogenic Keratocyst: A Systematic Review and Meta-analysis - Odontology

Manuscripts Under Preparation

- Pre-pandemic FT3, FT4, and TSH serum Levels as Independent Predictors of SARS-CoV-2 Infection Risk: Evidence from the Qatar Biobank
- Age-Specific Association Between Periodontitis and All-Cause Mortality in U.S. Adults: Findings from NHANES III
- Association Between Proton Pump Inhibitor Use and Denture Status Among Older Adults: A Cross-Sectional Analysis of NHANES 2005–2008
- Association Between Proton Pump Inhibitor Use and Periodontitis Severity Among U.S. Adults: A Cross-Sectional Analysis of NHANES 2009-2010
- Proton pump Inhibitors, Salivary composition and Oral health: A cross-sectional Study
- Scoping Review of Regulatory Approval for AI Medical Devices in Dentistry
- Scoping Review of Usability studies of AI based mobile applications
- AI-Assisted vs Unassisted Diagnostic Performance in Oral Lesion Detection: A Multi-Reader Study
- Artificial intelligence in pediatric dentistry – A Narrative review
- Artificial Intelligence Integration with Salivary Liquid Biopsy for Pre-Malignant Transformation Prediction: A Scoping Review
- Cognitive Bias in Diagnosing Apical Lesions on Panoramic Radiography: A Comparative Accuracy Study with CBCT

BOOKS & BOOK CHAPTERS

- Rishal M Pathuthara, Osama ElKashty, Dipak Sapkota, Akhilanand Chaurasia, Faleh Tamimi. ***The role of artificial intelligence in advancing digital pathology for oncology***: Book Chapter: AI and Precision Oncology, Elsevier Publication [Manuscript Under Review]
- Future of Ai Healthcare and hospitals: Dentomed Publishing House.

RESEARCH PROPOSALS

- Diagnosis of Odontogenic Infections using Deep learning
- Deployment of Artificial intelligence app for early diagnosis of Oral Cancer: a prospective Cohort study

- Design and Usability Testing of 'Dentech': An AI-Powered web platform for Screening of Oral Cancer and Potentially Malignant Disorders by Healthcare Professionals
- CIHR -Project Grant Application
- CIHR -Planning and Dissemination Grant application 2026

CONFERENCES & PRESENTATIONS

- AI and Medicine – Future Is Now, Sidra Medicine, 2025
- CDEM Clinical Conference, Qatar University – 2023, 2024, 2025
- International Association for Dental Research (IADR), Qatar – 2025
- Additional national and international conferences and workshops

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ACADEMIC SERVICE & OUTREACH

- Mentorship of undergraduate and international students through structured research internship programs (SRIP, IDRP).
- Organization and supervision of community dental outreach and oral cancer awareness programs.

REFERENCES

Available upon request